

ENTERPRISE JAVA			
Course Code	25MCA201	Course Type	PCC
Teaching Hours/Week (L:T:P:S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisites	Basic Programming Knowledge		
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	To understand Input/Output using Java and use Class fundamentals, Objects, Methods, Polymorphism,		
2.	To understand Inheritance, Abstraction, Package creation and Exceptions handling in Java		
3.	To understand Multithreaded Programming concepts using Java, develop GUI Applications in Java using Swing packages and Event handling.		
4.	To understand and use Networking concepts, Distributed computing using JDBC for Database connectivity.		
5.	To understand Web application development, Using Servlets, Spring Boot.		
UNIT - I			
			10 Hours
Introduction to JAVA :			
Introducing classes: Class fundamentals; Declaring objects; Assigning object reference variables; Introducing methods; Constructors; The this keyword; Overloading methods; Introducing access control; Understanding static: Introducing final; introducing nested and inner classes.			
Stream Classes:			
Reading and writing files; Stream classes; Byte Streams (InputStream; OutputStream; FileInputStream; FileOutputStream; DataInputStream, DataOutputStream); Character streams(Reader; Writer, FileReader, FileWriter, BufferedReader, BufferedWriter); Serialization: Serializable, ObjectOutputStream, ObjectInputStream.			
UNIT -II			
			06 Hours
Object Oriented Fundamentals and Exception Handling			
Inheritance :			
Inheritance basics; Using super; Creating a multilevel hierarchy; Method overriding; Dynamic method dispatch; Using Abstract class; Using final with inheritance, Aggregation, Packages; Access protection; Importing packages; Interfaces.			
Exception handling :			
Exception- handling fundamentals; Exception types; Uncaught exceptions; Using try and catch; Multiple catch clauses; Nested try statements; throw; throws; finally; Java’s built-in exceptions; Creating your own exception sub-classes;			
UNIT - III			
			10 Hours
The Collection Framework, Multithreaded Programming and Event Handling, AWT and Swings:			
The Collection Framework:			
The Collection interfaces: The Collection interface, The List interface, The Set interface; The collection classes: The ArrayList class, The LinkedList class, The Hashset class, The Linked HashSet class.			
Multithreaded programming:			

Java Thread model; Main thread; Creating a thread; Creating multiple threads; Using isAlive() and join(); Synchronization; Interthread communication

Event Handling, AWT and Swings :

Event Handling:

Two event handling mechanisms; Delegation event model; Event classes; Sources of events; EventListener interfaces; Using the Delegation Event Model

Swings :

JApplet; JFrame; Labels; Text fields; Buttons; Check boxes; RadioButtons; Adapter classes

Unit - IV

06 Hours

Networking , Database Connectivity, Java Beans:

RMI :

Remote Method Invocation (RMI): Remote Method Invocation concept; Server side; Client side.

Networking basics: Java and the net; InetAddress; TCP/IP client sockets; URL: URLConnection; TCP/IP server sockets; Datagrams.

DataBase Connectivity:

JDBC objects: Concept of JDBC; JDBC packages; Brief overview of the JDBC process; Database connection; Associating the JDBC/ODBC bridge with the database; Statement objects; ResultSet.

Java Beans :

Introduction to Java Beans; Advantages of Java Beans; JAR files; Developing a simple Bean; Using bound properties; Using BeanInfo Interface; Introduction to Constrained properties.

Unit - V

08 Hours

Servlets and Spring Boot :

Servlets :

Background; Life cycle of a Servlet; Simple Servlet; Servlet API; javax.servlet package; Reading Servlet parameter; javax.servlet.http package; Handling HTTP requests and responses; Using Cookies and sessions.

JSP:

JSP elements, Request and Response, Introduction to JSP implicit objects

Spring Boot :

Introduction to Spring Boot, Essential Key Components of spring Boot, Setting up a Spring Boot workspace, Developing first spring boot application.

Course Outcomes: At the end of the course student will be able to

1.	Analyze and Apply Input/Output operations in Java, leveraging Class fundamentals, Objects, Methods, and Polymorphism to design efficient and reusable software solutions.
2.	Evaluate and Implement advanced Object-Oriented Programming principles, including Inheritance, Abstraction, Package creation, and Exception handling, to enhance code modularity and robustness.
3.	Design and Develop multithreaded applications in Java, integrating GUI components using Swing and applying event-driven programming for responsive user interfaces.
4.	Apply and Implement networking concepts and database connectivity using JDBC to develop secure and efficient data-driven applications.
5.	Develop and Enhance web applications using Servlets and Spring Boot, applying best practices in web development and software architecture.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

	Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
	↓ Course Outcomes									1	2
	25MCA201.1	3	3	3	3	3	3	-	3	3	-
	25MCA201.2	3	3	3	3	2	2	-	3	3	-
	25MCA201.3	3	3	3	3	2	2	-	3	3	-
	25MCA201.4	3	2	2	2	2	3	-	3	3	-
	25MCA201.5	3	3	3	2	3	3	-	3	3	-
1: Low 2: Medium 3: High											
TEXT BOOKS:											
1.	Herbert Schildt : Java: The Complete Reference, 12 th Edition, Tata McGraw Hill., 2019										
2.	Mastering Spring Boot 2.0: Build modern, cloud-native, and distributed systems using Spring Boot, Dinesh Rajput, Pack Publishing Ltd., 2018.										
REFERENCE BOOKS:											
1.	Dr. Donald Doherty and Rick Leinecker : JavaBeans Unleashed										
2.	James Goodwill : Developing Java Servlets										
3.	Steven Holzner : Java 2 Black Book										
4.	Ed Roman : Mastering Enterprise JavaBeans										
5.	Jim Keogh : The Complete Reference J2EE, Tata McGraw Hill, 2008.										

ADVANCED WEB TECHNOLOGIES			
Course Code	25MCA202	Course Type	PCC
Teaching Hours/Week (L:T:P:S)	3-0-0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisite	Fundamentals of Programming		
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	To understand and apply the principles of XHTML, CSS, HTML5, and Bootstrap and develop the skills to create structured web pages, implement effective styling, leverage the features of HTML5 for modern web development, and utilize Bootstrap for responsive and visually appealing design, and overview of XML key components,		
2.	To understand PHP scripting language and its applications in web development.		
3.	To understand JavaScript and its programming concepts, and. JavaScript execution environment and event handler techniques		
4	To understand the JQuery concept and its effects and provide an introduction to AngularJS.		
5.	To develop an ability to design and implement static and dynamic website and client-server architecture to develop a web application		
UNIT – I			
XHTML:			02 Hours
Standard Structure, Basic Text Markup, Lists, Tables, Forms, Frames.			
CSS:			02 Hours
Introduction, Levels of Style Sheets, Selector Forms, Property Value Forms, Font Properties, List Properties, Color, Alignment of Text, The Box Model, The and <div> tags.			
HTML5:			02 Hours
Introduction of html5, iframes, layout, responsive web design(view port),computer code, new semantics elements in HTML5, Migration, canvas, svg, input types, new form elements and attribute, google map, media(audio, video)			
BOOTSTRAP:			02 Hours
Introduction of Bootstrap, First Web Page with Bootstrap, Bootstrap Grid system, CDN, Tables, Images, Jumbotron and Page Header, Wells, Alerts, Buttons, Badges and Labels, Progress Bars, List Groups, Navigation Bar, Forms, Form Inputs, Media Objects, Carousel Plug-in			
UNIT –II			
PHP:			04 Hours
Overview of PHP and uses of PHP, General Syntactic Characteristics, Primitives, Operations and expressions, Control Statements, Arrays, Functions, Pattern Matching, Form Handling, Files.			
Advanced PHP:			04 Hours
Cookies, Session tracking, Database Access with PHP and MySQL.			

UNIT – III	
JavaScript	04 Hours
Overview of JavaScript, Object Orientation and JavaScript, Syntactic Characteristics, Screen Output and Keyboard Input, Control Statements, Object Creation and Modification, Arrays, Functions, Constructors, Pattern Matching.	
Dynamic Object Model (DOM)	04 Hours
JavaScript Execution Environment, The Document Object Model, Element Access in JavaScript, Events and Event Handling, Handling Events from the Body Elements, Button Elements, Text Box and Password Elements, The Navigator Object.	
UNIT – IV	
Dynamic Documents with Java Script	04 Hours
Introduction to dynamic documents, Positioning Element, Moving Elements, Element Visibility, Changing Colors and Fonts, Dynamic Content, Stacking Elements, Locating the Mouse Cursor, Reacting to a Mouse Click, Slow Movement of Elements, Dragging and Dropping Elements.	
JQUERY:	02 Hours
Introduction, Syntax, selectors, events, JQuery HTML, Effects	
Introduction to Angular JS:	02 Hours
What is AngularJS? Angular Directives, AngularJS Expression, Angular Expression, AngularJS Module, AngularJS Controller, AngularJS Filter.	
UNIT – V	
Web 2.0 and Building Rich Internet Applications with AJAX	02 hours
Web 2.0 What is Web 2.0?, Folksonomies and Web 2.0, Software As a Service(SaaS), Data and Web 2.0, Convergence, Iterative development, Rich User experience, Multiple Delivery Channels, Social Networking. Web Services: SOAP, RPC Style SOAP, Document style SOAP, WSDL, REST services, JSON format, What is JSON?, Array literals, Object literals, Mixing literals, JSON Syntax, JSON Encoding and Decoding, JSON versus XML.	
Building Rich Internet Applications with AJAX	04 Hours
Building Rich Internet Applications with AJAX: Limitations of Classic Web application model, AJAX principles, Technologies behind AJAX, Examples of usage of AJAX, Dynamic web applications through Hidden frames for both GET and POST methods. IFrames, Asynchronous communication and AJAX application model, XMLHttpRequest Object – properties and methods, handling different browser implementations of XMLHttpRequest, The same origin policy, Cache control, AJAX Patterns (Only algorithms – examples not required): Predictive fetch pattern, Submission throttling pattern, Periodic refresh, Multi stage download.	
Building Advanced Web 2.0 applications	02 Hours
Definition of mash up applications, Mash up Techniques, Building a simple mash up application with AJAX,	
Course Outcomes: At the end of the course student will be able to	
1.	Create well-structured web pages using XHTML, apply CSS styling techniques, utilize HTML5 features for modern web development, and design responsive websites using Bootstrap, demonstrating proficiency in web development with these technologies as well as gain a solid understating of XML

2.	Develop proficiency in PHP programming by demonstrating knowledge of its manipulate arrays, create functions, handle form data, manage files, utilize cookies and session tracking, and interact with MySQL databases.
3.	Develop proficiency in JavaScript programming and creating dynamic web documents.
4.	Effectively use jQuery to manipulate HTML elements and enabling to apply jQuery for dynamic web development and explore further possibilities with AngularJS.
5.	Learn advanced Web 2.0 applications and overall concepts of Rich Internet Applications with AJAX

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes									1	2
25MCA202.1	3	3	3	2	3	-	3	3	3	-
25MCA202.2	3	3	2	2	3	-	3	3	3	-
25MCA202.3	3	3	3	3	2	-	3	3	3	-
25MCA202.4	3	3	2	3	3	-	3	3	3	-
25MCA202.5	3	3	3	3	2	-	3	3	3	-

1: Low 2: Medium 3: High

TEXT BOOKS:

1.	Robert W. Sebesta: Programming with World Wide Web, 8 th Edition, Pearson Education, 2014.
2.	Eric Van derVlist et al: Professional Web 2.0 Programming, Wiley India, 2007.

REFERENCE BOOKS:

1.	Internet & World Wide Web: How to Program, 5 th Edition by M. Deitel et al.
2.	Chris Bates: Web Programming Building Internet Applications, 3 rd Edition, Wiley India, 2007.
3.	Xue Bai et al: The Web Warrior Guide to Web Programming, Cengage Learning, 2001
4.	Thomas A. Powel: Ajax The Complete Reference, McGraw Hill, 2008.
5.	Gottfried Vossen, Stephan Hagemann: Unleashing Web 2.0 From Concepts to Creativity, Elsevier, 2007.
6.	Steven Holzner : Ajax Bible Wiley India , 2007.
7.	Justin Gethland et al: A Web 2.0 primer Pragmatic Ajax, SPD Publications, 2006.
8.	Jake Spurlock: Bootstrap- Responsive Web Development, O'Reilly Media, 2014.

SOFTWARE ENGINEERING AND TESTING				
Course Code		25MCA203	Course Type	PCC
Teaching Hours/Week (L:T:P:S)		3:1:0:0	Credits	04
Total Teaching Hours		50	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications				
Course Objectives:				
1.	Provide an overview of Agile development techniques			
2.	Know about system models and software designs.			
3.	Introduce Advanced Software Engineering Concepts.			
4.	Develop Proficiency in Advanced Software Testing Techniques.			
5.	Provide Experience in Performance, Security, and Scalability Testing.			
UNIT - I				
Agile Software development:				04 Hours
Agile methods, Agile development techniques.				
Software requirements:				06 Hours
Functional and non-functional requirements; Requirements engineering Process, Requirement elicitation, Requirement specification, Requirement Validation.				
UNIT-II				
System models:				05 Hours
Context models; Interaction models; sequence diagram Structured methods; class diagram, generalization, Aggregation. Behavioural Models-Data driven modelling; Event driven modelling, Model driven engineering.				
Software Design:				05 Hours
Object-oriented design using the UML-System context and interactions, Architecture design, Object class identification, Design models.				
UNIT-III				
Advanced Software Engineering Concepts				10 Hours
Software Engineering Paradigms & Models, Introduction to DevOps, and Continuous Integration/Deployment (CI/CD), Software Architecture & Design Patterns, Microservices and Serverless Architectures				
UNIT - IV				
Advanced Software Testing Techniques				10 Hours
Software Testing Fundamentals & Testing Levels (Unit, Integration, System), Automated Testing Tools (Selenium, JUnit, TestNG, Cypress), Static & Dynamic Testing (Code Analysis, Mutation Testing), Model-Based Testing (MBT) & AI-Driven Testing.				
UNIT - V				
Performance, Security & Scalability Testing				10 Hours
Performance Testing (Load, Stress, Endurance, Volume), Security Testing (Penetration Testing, OWASP Top 10, Threat Modelling), Scalability Testing in Cloud Environments, Testing in DevOps				

(Shift-Left & Shift-Right Testing).

Course Outcomes: At the end of the course students will be able to

1.	Understand Agile development and Requirement Engineering Process.
2.	Apply the different System Models and Software Designs on software engineering.
3.	Understand and Apply Advanced Software Engineering Principles.
4.	Summarize the relevance of Software Testing Techniques.
5.	Perform Performance, Security, and Scalability Testing.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes									1	2
25MCA203.1	3	2	2	2	-	-	-	3	3	2
25MCA203.2	3	3	3	3	-	-	-	3	3	2
25MCA203.3	3	2	3	3	-	-	-	3	3	2
25MCA203.4	3	3	3	3	2	-	-	2	3	2
25MCA203.5	3	3	3	3	3	2	2	3	3	2

1: Low 2: Medium 3: High
TEXT BOOKS:

1.	Ian Somerville: Software Engineering, 10 th Edition, Pearson Education Ltd
2.	Paul C. Jorgensen, Software Testing: A Craftsman's Approach, 4 th Edition, CRC Press
3.	Elfriede Dustin, <i>Effective Software Testing: 50 Specific Ways to Improve Your Testing</i> , Addison-Wesley

REFERENCE BOOKS:

1.	William Perry: Effective Methods for Software Testing, 2 nd Edition, John Wiley.
2.	Bezier B: Software Testing Techniques, 2 nd Edition, Wan Nostrand Reinluold.
3.	Myers G J: The Art of Software Testing, John Wiley.
4.	Roger S. Pressman: Software Engineering – A Practitioners Approach, 7 th Edition, McGraw-Hill

DATA WAREHOUSING AND DATA MINING			
Course Code	25MCA204	Course Type	PCC
Teaching Hours/Week (L:T:P:S)	3:1:0:0	Credits	04
Total Teaching Hours	50	CIE + SEE Marks	50+50
Prerequisites	Advanced Database Systems Mathematical Foundation for Computer Applications		
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	Provide the student with an understanding of the concepts of data warehousing, its architecture and modelling		
2.	Understand the Knowledge Discovery Process and get the notion of Data Mining		
3.	Learn the Preprocessing techniques required for Data Mining		
4.	Learn the data mining tasks of Association Analysis, Classification and Clustering and also study their well-known techniques		
5.	Test real data sets using popular data mining tools		
UNIT – I			
Data Warehousing & modelling:			10 Hours
Basic Concepts: Data Warehousing: A multitier Architecture, Data warehouse models: Enterprise warehouse, Data mart and virtual warehouse, Extraction, Transformation and loading, Data Cube: A multidimensional data model, Stars, Snowflakes and Fact constellations: Schemas for multidimensional Data models, Dimensions: The role of concept Hierarchies, Measures: Their Categorization and computation, Typical OLAP Operations			
UNIT – II			
Introduction to Data mining:			10 Hours
Introduction, What is data mining, Motivating Challenges, The Origins of Data Mining, Data Mining Tasks Data: Types of Data - Attributes and Measurements, types of Data Sets – Record data, graph based data, ordered data, Data Quality, Data Preprocessing – aggregation, sampling, dimensionality reduction, feature subset selection, feature creation, discretization and binarization, variable transformation, Measures of Similarity and Dissimilarity – Euclidean distance measures, similarity measures for binary data, simple matching coefficient, Jaccard coefficient and cosine similarity			
UNIT – III			
Association Analysis:			10 Hours
Problem Definition, Frequent Item set Generation, Rule generation. Alternative Methods for Generating Frequent Item sets, FP Growth Algorithm, Subjective and Objective measures for Evaluating Association Patterns			
UNIT – IV			
Classification:			10 Hours
Preliminaries, General Approach to solving classification Problem, Decision Tree Induction, Rule Based Classifiers, Nearest Neighbor Classifiers, Bayesian Classifiers, Evaluating the performance of classifier.			

UNIT – V											
Clustering Analysis:										10 Hours	
Overview, K-Means, Agglomerative Hierarchical Clustering, DBSCAN, Density-based Clustering, Graph-based clustering, Cluster Evaluation- Classification oriented measures, Similarity oriented measures											
Course Outcomes: At the end of the course student will be able to											
1.	Articulate the architecture of Data Warehouses and compare design schemas										
2.	Apply various preprocessing tasks and define Data Mining										
3.	Understand Association rule mining and apply association algorithms										
4.	Learn to apply various classification techniques and algorithms and evaluate										
5.	Interpret and Understand various clustering techniques and algorithms										
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
Program Outcomes→		1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes											
25MCA204.1		3	3	3	3	3	3	1	1	2	3
25MCA204.2		3	3	3	3	1	1	1	3	2	3
25MCA204.3		3	3	3	3	3	1	1	3	2	3
25MCA204.4		3	3	3	3	3	1	1	3	2	3
25MCA204.5		3	3	3	3	3	1	1	3	2	3
1: Low 2: Medium 3: High											
TEXT BOOKS:											
1.	Pang-Ning Tan, Michael Steinbach, Vipin Kumar: Introduction to Data Mining, Pearson, Fourth impression, 2018.										
2.	Jiawei Han, Micheline Kamber, Jian Pei: Data Mining -Concepts and Techniques, 3 rd Edition, Morgan Kaufmann Publisher										
REFERENCE BOOKS:											
1.	Arun K. Pujari : Data Mining Techniques, 3 rd Edition, Universities Press, 2013.										
2.	Sam Anahory, Dennis Murray: Data Warehousing in the Real World, Pearson, 10 th Impression, 2012.										
3.	Michael J. Berry, Gordon S. Linoff: Mastering Data Mining, Wiley Edition, 3 rd Edition, 2012										
E Books / MOOCs/ NPTEL											
1.	NPTEL Course on Data Mining, IIT Kharagpur, 2024										

PROFESSIONAL COMMUNICATION SKILLS			
Course Code	25MCA205	Course Type	PCC/Practical
Teaching Hours/Week (L:T:P: S)	1:1:0:0	Credits	02
Total Teaching Hours	25	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	Comprehend the process of communication and enhance presentation abilities for professional and social contexts.		
2.	Cultivate interpersonal skills, time management, and stress management for personal and professional growth.		
3.	Gain proficiency in effective writing by understanding its dynamics and structuring content clearly.		
4.	Recognize ethical considerations and responsibilities in professional practice.		
UNIT - I			
			11 Hours
Communication Skills Communication, Methods of Communication Effective Communication Skills One-way and Two-way Communication, Interpersonal & Social Skills			
Effective Presentation Strategies Defining the purpose of presentation, preparation and understanding nuances of delivery Formal Presentation, Sensitivity towards multi-cultural workspaces			
UNIT - II			
			10 Hours
Writing Skills Application letters and Resume writing, Formal E-mails, Email etiquette, Job Cover Letter			
Self-Enhancement Types of Personality, Johari window and SWOT Analysis and Interpersonal Skills, General Etiquettes			
UNIT - III			
			04 Hours
Ethics and Professional Responsibilities Rights and responsibilities regarding intellectual property, Workplace rights and responsibilities, Responsibilities for environment and society			
Time Management and Stress Time Management, Identifying the Stress, Signs of Stress, Healthier ways to deal with Stress.			

Course Outcomes: At the end of the course student will be able to

1	An ability to communicate effectively.
2	A recognition of the need for and an ability to engage in lifelong learning.
3	An ability to function effectively as an individual/member/leader on multidisciplinary teams and understanding of professional and ethical responsibility.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes									1	2
25MCA205.1	-	-	-	3	2	-	2	2	2	2
25MCA205.2	-	-	-	-	-	-	2	3	2	2
25MCA205.3	2	2	2	-	3	-	3	3	2	2

1: Low 2: Medium 3: High

TEXT BOOKS:

1.	Meenakshi Raman and Sangeeta Sharma: Technical Communication - Principles and Practices, Oxford University Press 2004.
2.	George Reynolds: Ethics in Information Technology, Thomson Course Technology, 2003.
3.	Ray French: Cross Culture Management, Universities Press
4.	Urmila Rai: Business Communication, Himalaya Publishing House
5.	Gerard Assey: Professional Business Etiquette & Grooming, https://www.barnesandnoble.com/w/professional-business-etiquette-grooming-gerard-assey eBook

REFERENCE BOOKS:

1.	M. Ashraf Rizvi: Effective Technical Communication by, 1 st Edition, Tata McGraw Hill.
2.	Mike W Martin and Ronald Scherzinger: Ethics in Engineering, 3 rd Edition Tata McGraw Hill.
3.	C.S Rayudu: Communication, Himalaya Publishing House
4.	Ronald B Adler & Jeanne Marquardt Elmhurst; Communicating at work – Principles and Practices for Business and the Professions McGraw-Hill College; Sixth Edition.
5.	Stephen P Robbins, Organizational Behaviour, Prentice Hall, India.
6.	Fred Luthans; Organizational Behaviour McGraw Hill International Edition. Carnegie, D. How to Win Friends and Influence People, New York: Gallery Books, 2019.
7.	Gaur, R. R., Sangal, R., & Bagaria, G. P. “Human Values and Professional Ethics”, New Delhi, India: Excel Books, 2010.
8.	Sarvesh Gulati: Corporate Grooming and Etiquette, Rupa Publisher 2016
9.	Dale Carnegie: Quick & Easy Way To Effective Speaking, Simon & Schuster Publication, 1990.

FULL STACK DEVELOPMENT USING DJANGO			
Course Code	25MCA211	Course Type	PEC
Teaching Hours/Week (L:T:P: S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	To introduce students to the Django web framework and its Model-View-Template (MVT) architecture.		
2.	To enable students to design and manage databases using Django’s Object-Relational Mapping (ORM).		
3.	To provide hands-on experience in creating dynamic web pages using Django templates and forms.		
4.	To develop skills in implementing user authentication, session management, and security best practices.		
5.	To familiarize students with REST APIs, AJAX-based interactivity, and deployment of Django applications on cloud platforms.		
UNIT - I			
Django Basics & MVT Architecture:			08 Hours
What is a Web Framework? Introduction to Django, MVC vs MVT Architecture, Creating and Importance of virtual environments (In VS Code), Installing and setting up Django projects and apps, Project structure explained (settings.py, urls.py, views.py), Creating views and mapping URLs, HTTP request-response cycle in Django, Debugging with error pages, Concept of loose coupling in Django, Using virtual environments and pip, Working with development vs production settings			
UNIT - II			
Models, ORM & Database Integration :			08 Hours
Introduction to Django ORM and relational databases, Creating models and fields, Model relationships (OneToOne, OneToMany, ManyToMany), Making migrations and applying them, The QuerySet API: creating, retrieving, filtering, updating, deleting records, Adding __str__, Meta, ordering, and verbose_name, Understanding model managers, Schema evolution and database migrations, Using PostgreSQL, MySQL, and SQLite with Django, Custom model methods			
UNIT - III			
Django Templates, Static & Forms :			08 Hours
Template system basics and directory structure, Template inheritance and blocks, Template tags, filters, and context processors, Working with static files (CSS, JS) and media files (uploads), Creating Django forms (Form, ModelForm), Custom validation (clean methods), Rendering forms manually vs {{form.as_p }}, Manual vs auto form rendering, CSRF protection in Django, Built-in form field types and widgets, Integrating Bootstrap with templates			
Admin Interface, CBVs, State Management :			08 Hours
Activating and using the Django admin site, Customizing model admin (list_display, search_fields, readonly_fields), Introduction to Function-Based Views (FBV) and Class-Based Views (CBV), Using ListView, DetailView, CreateView, UpdateView, DeleteView, Form handling using CBVs, Sessions, cookies, and state management. Authentication system: login, logout, signup, password			

reset, Permissions, groups, and decorators (login_required), Handling file uploads (images, PDFs), Overriding default user model (basic introduction)

APIs, AJAX, Security & Deployment :
08 Hours

Introduction to REST APIs and serializers (conceptual), Using fetch and axios for frontend integration, AJAX basics and jQuery in Django, Sending/receiving JSON data in views, CSRF and CORS in AJAX, Generating and downloading CSV and PDF files, Introduction to deployment on PythonAnywhere/Heroku, Django project structure for production, Django settings for production (debug, allowed hosts, static files), Security best practices: XSS, SQL Injection, clickjacking prevention

Course Outcomes: At the end of the course student will be able to

1	Understand Django's Architecture and Core Concepts
2	Develop Models and Integrate Databases using Django ORM
3	Design Dynamic Web Pages Using Templates and Forms
4	Implement Secure User Authentication and State Management
5	Integrate Frontend Logic, Generate Non-HTML Output and Deploy Projects

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes									1	2
25MCA211.1	3	2	--	--	--	--	--	--	3	--
25MCA211.2	3	3	2	2	--	--	--	--	3	2
25MCA211.3	2	3	3	2	2	--	--	--	3	3
25MCA211.4	2	2	3	3	2	2	--	--	3	2
25MCA211.5	3	2	3	3	3	3	2	2	3	3

1: Low 2: Medium 3: High

TEXT BOOKS:

1. Adrian Holovaty, Jacob Kaplan Moss, The Definitive Guide to Django

REFERENCE BOOKS:

1. William Vincent – Django for Beginners
2. Antonio Mele – Django 4 by Example
3. Arun Ravindran – Django Design Patterns and Best Practices

WEB DEVELOPMENT USING RUBY ON RAILS & POSTGRESQL			
Course Code:	25MCA212	Course Type	PEC
Teaching Hours/Week (L: T: P: S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisite			
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	Learn how websites work and how Ruby on Rails helps build them.		
2.	Build web pages and connect them using routes, controllers, and views		
3.	Understand how data is stored and managed using PostgreSQL		
4.	Connect Rails apps with PostgreSQL to save and use data		
5.	Learn how to deploy web apps online and keep them running		
UNIT-I			
Web Fundamentals & Getting Started with Rails			08 Hours
Basics of web architecture (HTTP, request/response cycle), MVC architecture in Rails, Introduction to Ruby, Rails project structure			
Installing Rails and PostgreSQL, Creating first Rails project, Scaffold generation, Running server			
UNIT-II			
Routing, Controllers, and Views			08 Hours
Routing in Rails, Controllers and actions, View rendering with ERB, Flash messages, Layout templates			
Dynamic pages, HTML forms in Rails, Navigation, Handling user input			
UNIT-III			
Models, Associations & User Management			08 Hours
ActiveRecord ORM, Migrations, Validations and Callbacks, Model Relationships, Devise Authentication.			
Creating models, Form validations, Sign-up/Login, Simple blog with users and posts			
UNIT-IV			
PostgreSQL Basics and SQL Queries			08 Hours
PostgreSQL architecture, Data types, SQL basics (SELECT, INSERT, etc.), Joins, Indexing.			
Creating/querying databases, SQL commands, Using pgAdmin/CLI, Fetching related data			
UNIT-V			
PostgreSQL in Rails & Deployment Concepts			08 Hours
Connecting Rails to PostgreSQL, Schema management, Indexes, Transactions, Deployment (Heroku)			
Rails-PostgreSQL integration, Migrations, Seed data, Deploy to Heroku			
Course Outcomes: At the end of the course student will be able to			
1.	Describe the structure of a web application and explain the role of Ruby on Rails in web development.		
2.	Develop basic web applications using controllers, views, models, and routes in Rails.		
3.	Write SQL queries to create, retrieve, update, and delete data using PostgreSQL.		
4.	Integrate a PostgreSQL database with a Rails application for data-driven features		
5.	Deploy a full-stack web application to a live server (like Heroku) with database support.		

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
	Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
	↓ Course Outcomes									1	2
	25MCA212.1	3	--	--	2	--	--	--	2	3	2
	25MCA212.2	3	2	3	3	2	--	--	2	2	2
	25MCA212.3	3	2	--	2	--	--	--	2	2	2
	25MCA212.4	3	3	2	3	2	--	--	2	2	3
	25MCA212.5	--	2	3	3	3	2	2	2	2	2
1: Low 2: Medium 3: High											
TEXTBOOKS:											
1.	Agile Web Development with Rails 7 by Sam Ruby, Pragmatic Bookshelf (2023), ISBN: 9781680509293										
2.	The Odin Project – Full Stack Ruby on Rails Path (Free, Online) URL: https://www.theodinproject.com/paths/full-stack-ruby-on-rails										
3.	PostgreSQL: Up and Running by Regina O. Obe & Leo S. Hsu, O'Reilly Media, 3rd Ed (2021), ISBN: 9781491963418										
4.	Learning PostgreSQL 11 by Salahaldin Juba and Andrey Volkov, Packt Publishing (2019), ISBN: 9781789535464										
REFERENCE BOOKS:											
1.	The Rails 5 Way, Obie Fernandez, Addison-Wesley Professional (2017), 9780134657674, Deep dive into professional Rails programming practices										
2.	Ruby on Rails Tutorial: Learn Web Development with Rails (6th Edition), Michael Hartl, Addison-Wesley Professional (2022), ISBN: 9780138051836 Step-by-step tutorial ideal for beginners, includes testing, authentication, and deployment										
3.	Learning PostgreSQL 11, Salahaldin Juba, Andrey Volkov. Packt Publishing (2019) ISBN: 9781789535464										
4.	Mastering PostgreSQL in Application Development, Dimitri Fontaine, The SQL Company (2019), ISBN:9782956601402										
E Books / MOOCs/ NPTEL											
1.	NPTEL Course: Programming, Data Structures and Algorithms using Python, By Prof. Madhavan Mukund IIT Madras.										
2.	MOOC: Ruby on Rails Web Development Specialization, Johns Hopkins University, Coursera, 2023.										

APPLIED DATA SCIENCE AND ANALYTICS			
Course Code:	25MCA221	Course Type	PEC
Teaching Hours/Week (L:T:P:S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	Understand the overview of the Data Science Process.		
2.	Apply the knowledge of Data Collection and Preprocessing to acquire the data and clean the data for further training.		
3.	Understand how to perform Exploratory Data Analysis using matplotlib, pandas and Seaborn.		
4.	Use Statistical Inference to gain key information from the dataset.		
5.	Use machine learning to create a data model and evaluate the model w.r.t its accuracy, precision and recall.		
UNIT - I			
Introduction to Data Science			04 Hours
What is Data Science? Data Science vs. Data Analytics vs. Business Intelligence, Data Science in the Big Data World, Types of Data- Structured, Unstructured, Natural Language, Machine Generated, Graph Based, Audio, Videos, Images & Streaming Data, Quantitative Vs Categorical Data.			
The Data Science Process- Defining Research Goals, Retrieving Data, Cleansing, integrating and transforming data, Exploratory Analysis, Build the models, presenting findings and building applications.			
Python for Data Science:			04 Hours
Data Types-int, float, bool, str, list, tuple, dictionary, set, Functions. Exploring NumPy Library- Indexing, Slicing, One dimensional Array, Multi-Dimensional Array, Concatenation and Splitting Arrays, Arithmetic operations, Aggregation, Sorting, Partitioning.			
UNIT - II			
Data Collection and Preprocessing			04 Hours
Getting Data: Reading Files, Scraping from Web, Using API's. Data cleaning: missing values, duplicates, encoding, scaling.			
Data Manipulation using Pandas			04 Hours
Loading Data- read_csv, read_excel, read_html. Data Cleaning and Preparation- Filtering out missing data, Filling in Missing Data, Removing Duplicates, Replacing Values, Detecting and filtering Outliers, String Manipulation.			
UNIT - III			
Exploratory Data Analysis (EDA)			04 Hours
Elements of Structured data, Estimates of Location-Mean, Median, Mode, Outliers, Estimates of Variability- Standard Deviation, Percentiles, Boxplots, Frequency Tables and Histograms, Density Plots. Exploring Binary and Categorical data- Bar Charts, Pie Charts. Correlation, Scatterplots			
Plotting using Matplotlib, Pandas & Seaborn			04 Hours
Line plots, Scatter Plots, Density and Contour plots, Histogram, Bar plots, Corelation Matrix, Heat Maps.			

UNIT - IV											
Statistical inference for Data Science										04 Hours	
Data and Sampling Distributions - Random Sampling and Sample Bias, Sample Mean Vs Population Mean, Selection Bias, Bootstrapping, Resampling Vs Bootstrapping. A/B Testing, Hypothesis Tests.											
Data Analysis using Python										04 Hours	
Perform Data analysis on Iris and Titanic Dataset.											
UNIT - V											
Data Modelling & Data Ethics										04 Hours	
What is a model? What is Machine Learning? Supervised, Unsupervised and Reinforcement Learning, Overfitting and Underfitting, Model Evaluation -Accuracy, Precision & Recall, The Bias-Variance Trade-off.											
What is Data Ethics, Building bad data products, Trading off accuracy and fairness, Recommendations, Biased Data, Data Protection											
Data Modelling using Python										04 Hours	
Perform Data Modelling on Titanic Dataset using Logistic Regression and evaluate the model.											
Course Outcomes: At the end of the course student will be able to											
1.	Describe the concept of the Data Science Process.										
2.	Apply the knowledge of Data Collection and Preprocessing to prepare the dataset for further training.										
3.	Perform Exploratory Data Analysis to identify critical information from data for further analysis.										
4.	Apply Statistical inference to back the data science model.										
5.	Create data models and evaluate them for their accuracy, precision and recall.										
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
	Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
	↓ Course Outcomes									1	2
	25MCA221.1	3	3	3	3	-	-	-	3	3	3
	25MCA221.2	3	3	3	3	-	-	-	3	3	3
	25MCA221.3	3	3	3	3	-	-	-	3	3	3
	25MCA221.4	3	3	3	3	-	-	-	3	3	3
	25MCA221.5	3	3	3	3	-	-	3	3	3	3
1: Low 2: Medium 3: High											
TEXT BOOKS:											
1.	Introducing Data Science- Big data, machine learning, and more, using Python tools, Davy Cielen, Arno D. B. Meysman, and Mohamed Ali, Manning Publications, 2016.										
2.	Practical Statistics for Data Scientists, by Peter Bruce, Andrew Bruce, O'Reilly Media, Inc., May 2017										
3.	Python Data Science Handbook, by Jake VanderPlas, O'Reilly Media, Inc., 2nd Edition										
REFERENCE BOOKS:											
1.	Python for Data Analysis, by Wes McKinney, O'Reilly Media, Inc., 3rd Edition										
2.	Data Science from Scratch, by Joel Grus, O'Reilly Media, Inc., 2nd Edition.										
3.	Doing Data Science, by Cathy O'Neil, Rachel Schutt, O'Reilly Media, Inc.										
E Books / MOOCs/ NPTEL											
1.	Python for Data Science - By Prof. Ragunathan Rengasamy IIT Madras NPTEL, 2025										

DIGITAL IMAGE PROCESSING & PATTERN RECOGNITION			
Course Code	25MCA222	Course Type	PEC
Teaching Hours/Week (L: T: P: S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisite	Data Structures with Algorithms Mathematical Foundation for Computer Applications		
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	To understand the fundamentals of digital image processing.		
2.	To implement spatial and frequency domain techniques for image enhancement.		
3.	To apply image restoration and segmentation techniques.		
4.	To understand various pattern recognition methods.		
5.	To implement different image classification Clustering techniques and to develop real-world applications using image processing and pattern recognition		
UNIT-I			
Fundamentals of Digital Image Processing			03 Hours
Introduction to Digital Image processing, Elements of visual perception, Image formation & human vision system; Applications in various domains (medical, agriculture, security, etc.)			
Image Representation & Basic Operations			04 Hours
Digital image representation; Image acquisition, sampling & quantization; Image transformations.			
UNIT-II			
Image Enhancement in Spatial Domain			04 Hours
Histogram processing; Point processing; Spatial filtering: smoothing (mean, median, Gaussian); Sharpening (Laplacian, high-boost)			
Image Enhancement in Frequency Domain			04 Hours
Fourier Transform (DFT, FFT); filtering in frequency domain; Low-pass & high-pass filtering (Gaussian, Butterworth filters)			
UNIT-III			
Image restoration and Reconstruction			04 Hours
Noise models, restoration in the presence of noise, inverse filtering.			
Morphological Image Processing			03 Hours
Preliminaries, erosion and dilation, opening and closing, basic morphological algorithms.			
Image Segmentation			03 Hours
Fundamentals, point, line and edge detection, thresholding, region-based segmentation.			
UNIT-IV			
Pattern Recognition Techniques			04 Hours
Introduction to Pattern Recognition; Supervised vs. unsupervised learning; Feature space & decision boundaries			
Statistical & Probabilistic Models			03 Hours
Bayes decision theory; Naïve Bayes classifier; Linear Discriminant Analysis (LDA)			

UNIT-V											
Image Classification										04 Hours	
Principal Component Analysis (PCA); Support Vector Machines (SVM); K-Nearest Neighbors (KNN).											
Clustering										04 Hours	
Overview of clustering; Different distance functions and similarity measures, K-means clustering; Hierarchical clustering; DBSCAN.											
Course Outcomes: At the end of the course student will be able to											
1.	Apply fundamental concepts of digital image processing to explain the image representation.										
2.	Implement spatial and frequency domain techniques for image enhancement.										
3.	Perform image restoration and segmentation for feature extraction.										
4.	Analyze various pattern recognition techniques and their applications.										
5.	Implement different image classification Clustering techniques. Develop real-world applications of digital image processing and pattern recognition.										
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
	Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
	↓ Course Outcomes									1	2
	25MCA222.1	3	3	3	3	3	2	2	2	3	3
	25MCA222.2	3	3	3	3	3	2	2	2	3	3
	25MCA222.3	3	3	3	3	3	2	2	2	3	3
	25MCA222.4	3	3	3	3	3	2	2	2	3	3
	25MCA222.5	3	3	3	3	3	2	2	2	3	3
1: Low 2: Medium 3: High											
TEXT BOOKS:											
1.	Rafael C. Gonzalez & Richard E. Woods : Digital Image Processing, 4th Edition, PHI										
2.	R.O. Duda, P.E. Hart and D.G. Stork, Pattern Classification, John Wiley.										
REFERENCE BOOKS:											
1.	Anil K. Jain : Fundamentals of Digital Image Processing, Prentice Hall.										
2.	William K. Pratt, John Wiley & Sons Inc : Digital Image Processing, 3 rd Edition.										
3.	Statistical Pattern Recognition; K. Fukunaga; Academic Press.										
4.	S. Theodoridis and K. Koutroumbas, Pattern Recognition, 4 th Edition, Academic Press.										
5.	Robert Schalkoff, Pattern Recognition: Statistical, Structural and Neural Approaches, John Wiley & Sons.										

MOBILE APPLICATION DEVELOPMENT USING ANDROID AND FLUTTER				
Course Code		25MCA231	Course Type	PEC
Teaching Hours/Week (L: T: P: S)		3-0-0	Credits	03
Total Teaching Hours		40	CIE + SEE Marks	50 + 50
Teaching Department: Master of Computer Applications				
Course Objectives: Students can learn mobile application development by understanding the following tools and components.				
1.	Understand the Fundamentals of Mobile Application Development.			
2.	Develop Android Applications Using Core Components.			
3.	Design Interactive User Interfaces and Implement Location-Based Features.			
4.	Implement Messaging, Networking, and Services in Android.			
5.	Build Cross-Platform Applications Using Flutter and iOS Tools.			
UNIT-I				
Introduction				8 Hours
Preliminary Considerations – Cost of Development, Importance of Mobile Strategies in the Business World, Effective use of Screen Real Estate, Understanding Mobile Applications Users, Understanding Mobile Information Design, Understanding Mobile Platforms, Using the Tools of Mobile Interface Design.				
UNIT-II				
Getting Started with Android Programming				8 Hours
What is Android? Obtaining the required tools, Anatomy of an Android Application, Components of Android Applications, Activities, Fragments, Utilizing the Action Bar.				
UNIT-III				
Android UI Design and Location Based Services				8 Hours
Views and View Groups, Basic Views, Displaying Maps, Getting Location Data, Preparing for Publishing, Deploying APK Files.				
UNIT-IV				
Android Messaging and Networking				8 Hours
SMS Messaging, Sending Email, Networking, Downloading Binary Data, Text files, Accessing Web Services, Performing Asynchronous Call, Creating your own services.				
UNIT-V				
Flutter and IOS				8 Hours
Flutter: Introduction, Installation, Creating Simple Application in Android Studio, Architecture Application, Introduction to Dart Programming, Widgets, Layouts, Gestures. IOS: Obtaining the tools and SDK, Components of XCODE, Architecture of iOS, Objective C and Swift Programming Languages				

Case Studies:

Design and develop a mobile application to display user profiles with 3 UI activities using intents, design an application that contains Phone Contacts in a vertical linear manner. Selected contact appears at the top of the list with a large, italicized font and a blue background, create an application that uses Layout Managers and Event Listeners, develop a mobile application to list the tourist places of Karnataka using List View, develop a mobile application to register a form in first activity and display the registered information in second activity using intents, develop a standard calculator application to perform basic calculations like addition, subtraction, multiplication and division, devise an application that draws basic graphical primitives (rectangle, circle) on the screen, build a mobile application that creates, saves, updates and deletes data in a database.

Course Outcomes: Upon completion of the course, the students will be able to:

1.	Illustrate effective user interfaces that leverage evolving mobile device capabilities.
2.	Develop applications using different SDK frameworks and toolkits.
3.	Establish various methods to integrate database and server-side technologies.
4.	Design and develop open-source software based mobile applications.
5.	Build and deploy competent mobile development solutions.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes									1	2
25MCA231.1	3	2	-	3	3	-	3	3	3	3
25MCA231.2	3	2	-	3	3	-	3	3	3	3
25MCA231.3	3	2	-	3	3	-	3	3	3	3
25MCA231.4	3	3	-	3	-	-	3	3	3	3
25MCA231.5	3	3	-	3	-	-	3	3	3	3

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	Jeff McWherter and Scott Gowell, "Professional Mobile Application Development", 1 st Edition, 2012.
2.	Wei-Meng Lee, "Beginning Android Application Development", Wiley 2011.

REFERENCE BOOKS:

1.	Reto Meier, "Professional Android 4 Application Development", Wrox Publications 2012.
2.	Rap Payne, "Beginning App Development with Flutter: Create Cross-Platform Mobile Apps", 1 st Edition, Publications 2019.

BLOCK CHAIN TECHNOLOGY			
Course Code	25MCA232	Course Type	PEC
Teaching Hours/Week (L: T: P:S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	To understand Block Chain- Blockchain Architecture and its Applications.		
2.	To understand crypto currencies- Types and Applications		
3.	To understand Concept of Double Spending- Hashing- Mining- payment verification- Resolving Conflicts and Creation of Blocks		
4.	To understand Crypto currency wallets and conversion to Fiat Currency.		
5.	To understand Smart contracts- usage- application- working principle- law and regulations.		
UNIT-I			
Introduction			08 Hours
Introduction to Blockchain, How Blockchain works, Blockchain vs Bitcoin, Practical applications, public and private key basics, pros and cons of Blockchain, Myths about Bitcoin.			
UNIT-II			
Blockchain Architecture			08 Hours
Blockchain Architecture, versions, variants, use cases, Life use cases of blockchain, Blockchain vs shared Database, Introduction to crypto currencies, Types, Applications.			
UNIT-III			
Double Spending			08 Hours
Concept of Double Spending, Hashing, Mining, Proof of work. Introduction to Merkel tree, Privacy, payment verification, Resolving Conflicts, Creation of Blocks.			
UNIT - IV			
Introduction to Bitcoin			08 Hours
Introduction to Bitcoin, key concepts of Bitcoin, Merits and De Merits Fork and Segwits, Sending and Receiving bitcoins, choosing bitcoin wallet, Converting Bitcoins to Fiat Currency.			
UNIT-V			
Introduction to Ethereum			08 Hours
Introduction to Ethereum, Advantages and Disadvantages, Ethereum vs Bitcoin, Introduction to Smart contracts, usage, application, working principle, Law and Regulations. Case Study			
Course Outcomes: At the end of the course student will be able to			
1.	Explain the core concepts of blockchain technology and differentiate it from traditional databases and systems.		
2.	Analyze blockchain architecture and assess its real-world applications, including the role of cryptocurrencies.		
3.	Demonstrate an understanding of security mechanisms like hashing, proof of work, and Merkle trees in blockchain.		
4.	Describe Bitcoin’s working principles, including transactions, wallets, and conversion to fiat currency.		
5.	Evaluate Ethereum and smart contracts by examining their applications, advantages, and legal considerations.		

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
	Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
	↓ Course Outcomes									1	2
	25MCA232.1	-	3	2	-	-	-	2	-	3	1
	25MCA232.2	-	3	2	-	-	-	2	-	3	1
	25MCA232.3	-	3	2	-	-	-	2	-	3	1
	25MCA232.4	-	3	2	-	-	-	2	-	3	1
	25MCA232.5	-	3	2	-	-	-	2	-	3	1
1: Low 2: Medium 3: High											
TEXTBOOKS:											
1.	Bikramaditya Singhal, Gautam Dhameja, Priyanshu Sekhar Panda, , “Beginning Blockchain: A Beginner's Guide to Building Blockchain Solutions”, APress, 2018 (Reprint year 2024)										
2.	Arshdeep Bahga, Vijay Madiseti, “Blockchain Applications: A Hands-On Approach”. 2018										
3.	Melanie Swan, “Blockchain- Blueprint for a new economy”, 2015, Oreilly										
4.	Daniel Drescher , “Blockchain Basics-A Non-Technical Introduction in 25 steps”, APress, Second Edition 2020.										
REFERENCE BOOKS:											
1.	Aravind Narayan. Joseph Bonneau, “Bitcoin and Crypto currency Technologies”, Princeton										

ENTERPRISE JAVA LAB													
Course Code				25MCA206			Course Type			PCC			
Teaching Hours/Week (L: T: P: S)				0:0:4:0			Credits			02			
Total Teaching Hours				26			CIE + SEE Marks			50+50			
Teaching Department: Master of Computer Applications													
Course Objectives:													
1.	To understand and implement Object Oriented Programming in Java, handling of exceptions and multithreaded programs.												
2.	To understand and implement network and distributed programming in Java												
3.	To understand Java APIs to connect to a backend database and development of Graphic User Interface for Java Applications.												
4.	To understand and implement Java components as well as develop web applications in Java.												
List of Experiments													
1.	Programs to implement encapsulation, abstraction, inheritance and polymorphism												
2.	Programs to handle Exceptions												
3.	Implementation of Multithreading Programs.												
4.	Implementation of Stream classes.												
5.	Network programming using Sockets and Datagrams												
6.	Implementation of Swing components												
7.	Programs to connect to backend databases using JDBC												
8.	Programs to Implement Java Beans.												
9.	Development of Server sided programs using servlets and JSP.												
10.	Implementation of Spring Boot 2.0												
Course Outcomes: At the end of the course student will be able to													
1.	Develop Java applications using Object Oriented Programming features, exception handling, multithreading and networking features.												
2.	Develop Java applications that connect to a backend database, provide Graphic User Interface, develop Custom components as well as web applications.												
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes													
	Program Outcomes→			1	2	3	4	5	6	7	8	PSO	PSO
	↓ Course Outcomes											1	2
	25MCA206.1			3	3	3	3	-	3	-	3	3	-
	25MCA206.2			3	3	3	3	-	3	-	3	3	-
1: Low 2: Medium 3: High													
REFERENCE BOOKS:													
1.	Herbert Schildt : Java: The Complete Reference, 11 th Edition, Tata McGraw Hill., 2019												
2.	Mastering Spring Boot 2.0: Build modern, cloud-native, and distributed systems using Spring Boot, Dinesh Rajput, Pack Publishing Ltd., 2018.												
3.	Dr. Donald Doherty and Rick Leinecker : JavaBeans Unleashed												
4.	James Goodwill : Developing Java Servlets												
5.	Steven Holzner : Java 2 Black Book												
6.	Ed Roman : Mastering Enterprise JavaBeans												
7.	Jim Keogh : The Complete Reference J2EE, Tata McGraw Hill, 2008.												

ADVANCED WEB TECHNOLOGIES LAB

Course Code	25MCA207	Course Type	PCC
Teaching Hours/Week (L: T: P: S)	0:0:4:0	Credits	02
Total Teaching Hours	26	CIE + SEE Marks	50+50
Prerequisite	Fundamentals of Programming		

Teaching Department: Master of Computer Applications

Course Objectives:

- | | |
|----|--|
| 1. | To develop an ability to design and implement static and dynamic website |
| 2. | Use appropriate client-side or Server-side applications |
| 3. | Develop basic programming skills using Javascript and JQuery |
| 4. | Learn the language of web : HTML,CSS, Bootstrap, PHP, JSON |

List of Experiments

- | | |
|-----|--|
| 1. | Develop and demonstrate XHTML document that illustrate the use of images, tables, links, formatting tools, lists, forms, frames and style sheets |
| 2. | Develop and demonstrate HTML5 document that illustrate the use of new semantics elements, Migration, canvas, svg, input types, new form elements and attribute, google map, media(audio, video). |
| 3. | Develop and demonstrate XHTML document that includes javascript for the following <ul style="list-style-type: none"> a. I/O statements b. Control statements c. Arrays d. Objects e. Functions f. Constructors g. Event Handling through... <ul style="list-style-type: none"> Body elements Button elements Text box and password elements h. Element accessing, positioning and moving h. Stacking and slow movements to elements h. Dragging and dropping the elements h. Element visibility |
| 4. | Develop and Demonstrate XHTML document that includes javascript for Pattern Matching Using Regular Expressions |
| 5. | Design a XML document to store information and display the XML document. |
| 6. | Execute simple programs using PHP, Create a Web Application using PHP |
| 7. | Database programs using PHP |
| 8. | Simple programs using JQuery, JSON and AJAX, Angular |
| 9. | AJAX applications using various GUI components, Implementation of “Shopping cart” application using AJAX |
| 10. | AJAX application to keep track of user data and retrieve the session data |
| 11. | AJAX application to verify client side and server side data |
| 12. | Simple Flex applications, Flex applications using actionsript. |
| 13. | Web applications that use RPC |
| 14. | Building RESTful Web Services with JAX-RS |
| 15. | Simple SOAP application. |

Course Outcomes: At the end of the course student will be able to

- | | |
|----|---|
| 1. | Design and implement static and dynamic websites with good aesthetic sense of designing and latest technical know-how's |
| 2. | Analyze and apply the role of languages like HTML, Bootstrap, CSS, XML, JavaScript, PHP and protocols in the workings of the web applications and web Services. |

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓Course Outcomes									1	2
25MCA207.1	3	3	3	3	2	-	3	3	3	-
25MCA207.2	3	3	2	2	3	-	3	3	3	-

1: Low 2: Medium 3: High

REFERENCE BOOKS:

- | | |
|----|---|
| 1. | Programming the World Wide Web" is the 8th edition, published by Pearson. |
| 2. | Internet & World Wide Web: How to Program" by Paul Deitel, Harvey Deitel, and Abbey Deitel. |

E Resources

- | | |
|----|---|
| 1. | https://onlinecourses.swayam2.ac.in/nou22_cs03/preview |
| 2. | https://onlinecourses.swayam2.ac.in/aic20_sp32/preview |

TECHNICAL SEMINAR AND REPORT WRITING													
Course Code				25MCA208				Course Type			Seminar		
Teaching Hours/Week (L: T: P: S)				0:0:0:2				Credits			02		
Total Teaching Hours				15				CIE + SEE Marks			50		
Teaching Department: Master of Computer Applications													
Course Objectives:													
1.	To learn to prepare for a seminar on a technical topic and present it effectively using overhead projectors and presentation software.												
2.	To learn to prepare a well formatted and well-organized report with appropriate content using report preparation software.												
The student will prepare for a seminar on a technical topic and present it. A panel consisting of faculty members will evaluate the presentation and marks will be provided. The student will submit a report on the same. For the preparation of the presentation the student will refer to technical papers from journals. The presentation will be prepared using LaTeX tool.													
Course Outcomes: At the end of the course student will be able to													
1.	Review a technical topic												
2.	Present effectively												
3.	Help the students face placement interviews.												
4.	Prepare a well-organized report with appropriate content												
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes													
	Program Outcomes→			1	2	3	4	5	6	7	8	PSO	PSO
	↓ Course Outcomes											1	2
	25MCA208.1			3	1	-	2	-	-	3	2	3	2
	25MCA208.2			3	1	-		-	-	-	2	3	2
	25MCA208.3			1	1	-	1	-	-	3	1	3	2
	25MCA208.4			-	-	-	-	-	-	-	-	3	2
1: Low 2: Medium 3: High													
REFERENCE BOOKS:													
1.	Naomi Karten, “Presentation Skills for Technical Professionals: Achieving Excellence (Soft Skills for It Professionals)”- 23 Sept. 2010												
2.	Fink, A., 2009. Conducting Research Literature Reviews: From the Internet to Paper. Sage Publications												
3.	Day, R .A., 1992. How to Write and Publish a Scientific Paper, Cambridge University Press.												

FULL STACK WEB DEVELOPMENT USING MERN			
Course Code	25MCA301	Course Type	PCC
Teaching Hours/Week (L: T: P: S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	Understand the MVC architectural pattern and its role in web development. Articulate what React is and why it is useful.		
2.	Utilize modern JavaScript features like ES6+ classes, modules, and arrow functions. Making use of controlled components, create and verify forms. Use a JavaScript package manager (either npm or Yarn)		
3.	Build reusable React components and implement state management effectively. Making HTTP requests to read or modify data.		
4.	Develop and integrate RESTful APIs using Express.js and MongoDB. Implement client-side routing using React Router.		
5.	Students able to create CRUD operations, routing mechanisms, and UI components.		
UNIT-I			
Introduction to the MERN stack:			04 Hours
Introduction, The MVC architectural Pattern. MERN Components React, Node.js, Express, MongoDB, Advantages and real-world applications of MERN Stack, understanding Isomorphic and types			
MERN Components			04 Hours
Advantages and real-world applications of MERN Stack. Understanding Isomorphic and types. Welcome to React- Obstacles and Roadblocks, Reacts Future			
UNIT-II			
Understanding React			04 Hours
Server setup, setting up a Development Environment -NVM, Node Js, Project, NPM, Express, Understanding JSX and it's compilation process, Separate Script File, Transform, Automate, React Library			
Web Server:			04 Hours
React Components-React classes, Composing components, passing data- using properties, property validation, using children's Dynamic, React Lifecycle Methods.			
UNIT-III			
Understanding React State:			04 Hours
React State – Setting state, Event handling, communicating from child to parent, Stateless components, Designing Components-state vs props			
React Components			02 Hours
component hierarchy communication, Introduction to React Context API.			
UNIT-IV			
Building RESTful APIs			05 Hours
REST , HTTP method as Actions, JSON, Express, Request Object, Response objects, Middleware, The list API , The create API, Using the LIST API, Using the Create API, Error Handling.			

Mongo DB:										05 Hours		
Mongo DB: Basics, Schema initialization , MongoDB Node.JS Driver, callbacks, promises, Reading from MongoDB, Writing to MongoDB												
UNIT-V												
Working with React Router and forms:										04 Hours		
Routing Techniques, Simple Routing, Route parameters, Route Query String, Programmatic Routing, Nested Routes, Browser history.												
Forms:										04 Hours		
Forms, Filter form, Get API, Edit page, Building UI Components and Forms, update API, Delete API.												
Course Outcomes: At the end of this course, the students will be able to:												
1.	Understand the concepts of MVC components and architecture and Implement node js, Express and MongoDB.											
2.	Use JSX transformation and automatic recompiling for better React development.											
3.	Develop React components and manage UI state effectively.											
4.	Implement State management and event handling in React applications.											
5.	Build Restful APIs, perform CRUD operations, and integrate UI components efficiently.											
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes												
Program Outcomes→			1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes											1	2
25MCA301.1			3	3	2	2	2	1	1	1	3	2
25MCA301.2			2	3	3	3	3	1	1	1	3	3
25MCA301.3			2	3	3	3	3	1	1	1	3	3
25MCA301.4			2	3	3	3	3	1	1	1	3	3
25MCA301.5			3	3	3	3	3	2	1	1	3	3
1: Low 2: Medium 3: High												
TEXTBOOKS:												
1.	Pro MERN Stack, Vasan Subramanian, 2019, ISBN-13(pbk):978-1-4842-2653-7											
2.	MERN Quick Start Guide, Eddy Wilson Iriarte Koroloiva, 2018, PACKT Publication, ISBN 978-1-78728-108-0											
REFERENCE BOOKS:												
1.	Full stack React, Anthony Accomazzo, Ari Lerner, Nate Murray, Clay Allsopp, David Gutman, and Tyler McGinnis, 2017.											